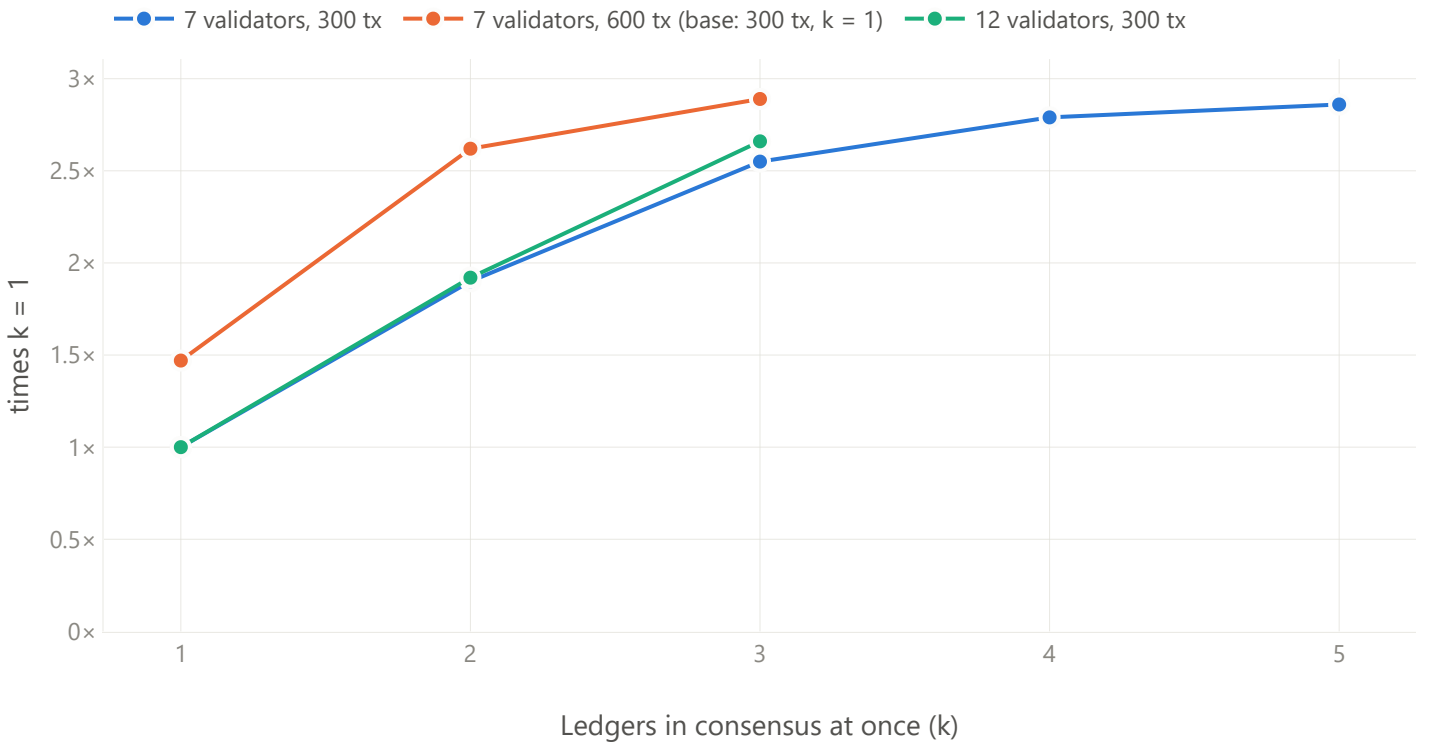Throughput relative to k = 1, mainnet apply cost